

Pattern	Dataset	Steps	Stage	AC	PPO	CSEAL	GEHRL-ST	GEHRL-EB	SRC	GRU4Rec	SASRec	UNO
Recent Learning	Junyi	5	trian	25.29 ± 0.58	25.73 ± 0.30	21.35 ± 0.16	21.77 ± 0.26	23.95 ± 0.49	11.58 ± 0.54	11.07 ± 4.93	27.26 ± 3.96	29.12 ± 0.21
			final	26.54 ± 0.58	25.25 ± 0.30	20.92 ± 0.16	22.25 ± 0.26	23.68 ± 0.49	8.23 ± 0.54	11.86 ± 4.93	23.04 ± 3.96	31.98 ± 0.21
			eval	28.79 ± 0.98	29.08 ± 0.47	25.58 ± 0.16	24.48 ± 0.86	25.88 ± 0.98	16.04 ± 3.31	16.88 ± 2.77	26.98 ± 5.15	34.78 ± 1.64
		10	trian	33.38 ± 0.13	33.59 ± 0.33	26.96 ± 1.09	29.30 ± 0.10	32.30 ± 0.47	11.71 ± 1.62	10.38 ± 4.30	28.69 ± 2.85	39.97 ± 2.61
			final	32.30 ± 0.13	32.70 ± 0.33	25.70 ± 1.09	26.98 ± 0.10	30.61 ± 0.47	7.82 ± 1.62	9.29 ± 4.30	28.54 ± 4.45	44.16 ± 2.61
			eval	35.84 ± 0.85	37.64 ± 1.01	30.19 ± 0.61	33.03 ± 1.29	34.89 ± 1.00	17.00 ± 2.64	11.67 ± 2.95	25.98 ± 3.91	46.01 ± 4.08
		20	trian	42.69 ± 0.13	42.94 ± 0.63	31.41 ± 0.08	37.73 ± 0.47	42.08 ± 0.65	10.99 ± 1.12	19.41 ± 10.15	38.65 ± 1.93	49.35 ± 1.02
			final	41.23 ± 0.13	41.94 ± 0.63	30.18 ± 0.08	37.34 ± 0.47	40.64 ± 0.65	6.44 ± 1.12	18.68 ± 10.15	34.69 ± 1.93	50.98 ± 1.02
			eval	45.71 ± 0.81	44.88 ± 1.42	34.21 ± 1.03	39.94 ± 2.38	44.16 ± 0.43	15.67 ± 0.80	20.18 ± 6.77	38.20 ± 3.65	54.05 ± 1.44
	Assist09	5	trian	27.64 ± 1.26	29.81 ± 0.53	26.07 ± 1.39	25.02 ± 0.15	26.19 ± 0.27	36.98 ± 4.30	21.12 ± 8.79	30.73 ± 0.96	38.31 ± 0.37
			final	25.02 ± 1.26	28.57 ± 0.53	22.99 ± 1.39	26.34 ± 0.15	24.31 ± 0.27	37.14 ± 4.30	14.66 ± 8.79	24.50 ± 0.96	40.13 ± 0.37
			eval	29.65 ± 1.18	30.50 ± 2.25	25.49 ± 1.74	25.52 ± 1.53	26.95 ± 0.79	38.90 ± 4.12	15.61 ± 11.49	23.63 ± 9.69	41.50 ± 0.21
		10	trian	30.18 ± 0.98	29.12 ± 0.24	28.29 ± 0.65	29.99 ± 0.71	30.86 ± 0.65	37.32 ± 3.81	27.83 ± 9.13	36.60 ± 3.14	39.57 ± 0.55
			final	26.86 ± 0.98	29.01 ± 0.24	25.97 ± 0.65	27.34 ± 0.71	30.41 ± 0.65	38.06 ± 3.81	24.00 ± 9.13	37.12 ± 3.14	40.40 ± 0.55
			eval	32.24 ± 1.40	29.81 ± 1.25	28.04 ± 2.40	30.36 ± 2.39	33.99 ± 0.39	39.66 ± 3.04	16.77 ± 17.60	38.74 ± 2.14	42.01 ± 0.20
		20	trian	30.46 ± 0.21	31.32 ± 0.61	29.75 ± 0.70	31.51 ± 0.57	31.72 ± 0.30	38.78 ± 1.78	16.95 ± 7.76	26.73 ± 6.16	40.48 ± 0.55
			final	29.46 ± 0.21	28.07 ± 0.61	24.72 ± 0.70	30.19 ± 0.57	30.70 ± 0.30	40.08 ± 1.78	19.85 ± 7.76	25.54 ± 6.16	40.73 ± 0.55
			eval	30.51 ± 2.48	30.94 ± 1.71	29.50 ± 1.90	33.91 ± 2.02	32.18 ± 1.64	41.82 ± 0.00	20.48 ± 13.28	30.71 ± 5.85	42.04 ± 0.41
Historical Revise	Junyi	10	trian	10.11 ± 0.59	9.29 ± 1.07	5.72 ± 0.40	7.73 ± 0.71	9.10 ± 0.71	-0.51 ± 1.96	-2.02 ± 4.91	9.29 ± 1.99	22.78 ± 2.75
			final	10.69 ± 0.59	10.18 ± 1.07	7.39 ± 0.40	9.23 ± 0.71	10.33 ± 0.71	-1.13 ± 1.96	4.64 ± 4.91	7.37 ± 1.99	23.35 ± 2.75
			eval	11.03 ± 1.10	10.46 ± 3.16	4.17 ± 1.09	6.98 ± 0.59	7.29 ± 0.93	-2.58 ± 3.20	1.21 ± 2.75	10.65 ± 2.88	29.19 ± 5.67
		20	trian	14.76 ± 0.47	16.52 ± 0.80	7.55 ± 0.54	11.19 ± 0.15	14.19 ± 0.32	-1.76 ± 1.04	1.50 ± 4.62	10.69 ± 3.13	27.62 ± 1.92
			final	10.01 ± 0.47	17.55 ± 0.80	2.99 ± 0.54	8.31 ± 0.15	10.14 ± 0.32	-10.72 ± 1.04	2.36 ± 4.62	3.77 ± 3.13	26.46 ± 1.92
			eval	15.55 ± 0.91	17.30 ± 3.95	10.20 ± 0.80	12.37 ± 1.44	16.18 ± 0.66	-7.57 ± 13.35	11.77 ± 2.56	8.18 ± 4.90	34.32 ± 2.48
		30	trian	19.89 ± 0.41	17.55 ± 0.48	9.74 ± 1.34	16.34 ± 0.66	19.06 ± 0.47	-4.16 ± 2.57	-4.55 ± 4.59	12.20 ± 3.18	30.03 ± 1.01
			final	20.88 ± 0.41	18.31 ± 0.48	10.55 ± 1.34	18.57 ± 0.66	18.99 ± 0.47	-1.57 ± 2.57	-5.11 ± 4.59	7.19 ± 3.18	28.00 ± 1.01
			eval	19.71 ± 0.94	19.37 ± 1.98	10.93 ± 2.65	16.37 ± 2.52	18.84 ± 1.29	-5.20 ± 7.73	-4.61 ± 8.29	14.63 ± 3.33	34.74 ± 3.88
	Assist09	10	trian	7.39 ± 3.31	3.28 ± 1.54	1.86 ± 3.79	5.59 ± 1.04	7.49 ± 0.63	28.50 ± 2.86	-3.87 ± 16.21	12.74 ± 6.72	32.46 ± 0.81
			final	-1.32 ± 3.31	-0.60 ± 1.54	-7.91 ± 3.79	2.11 ± 1.04	8.45 ± 0.63	27.04 ± 2.86	6.57 ± 16.21	7.98 ± 6.72	31.74 ± 0.81
			eval	8.51 ± 6.42	8.77 ± 1.14	-1.48 ± 6.96	8.27 ± 2.17	11.82 ± 2.96	30.37 ± 1.94	5.10 ± 23.80	8.24 ± 8.92	35.32 ± 0.59
		20	trian	9.63 ± 1.72	9.46 ± 4.17	4.59 ± 2.57	5.84 ± 0.69	9.60 ± 0.46	29.19 ± 2.28	10.22 ± 17.96	5.94 ± 12.04	32.41 ± 0.44
			final	4.62 ± 1.72	11.61 ± 4.17	2.41 ± 2.57	3.69 ± 0.69	11.04 ± 0.46	31.46 ± 2.28	5.88 ± 17.96	4.58 ± 12.04	34.14 ± 0.44
			eval	10.47 ± 1.52	17.96 ± 8.12	1.08 ± 2.26	11.19 ± 1.35	12.12 ± 2.11	34.85 ± 0.00	8.23 ± 34.68	8.56 ± 14.15	35.08 ± 0.02
		30	trian	10.89 ± 2.98	10.82 ± 2.43	-3.13 ± 1.32	6.94 ± 0.62	10.47 ± 1.05	24.08 ± 10.09	9.15 ± 16.59	4.35 ± 13.88	32.84 ± 0.43
			final	7.73 ± 2.98	13.49 ± 2.43	-3.09 ± 1.32	3.57 ± 0.62	10.95 ± 1.05	23.07 ± 10.09	5.53 ± 16.59	1.44 ± 13.88	31.65 ± 0.43
			eval	9.41 ± 3.24	17.22 ± 1.18	-2.25 ± 1.99	10.27 ± 2.75	11.95 ± 1.84	23.22 ± 16.45	9.71 ± 25.62	1.05 ± 22.69	35.15 ± 0.22
Exploratory Learning	Junyi	10	trian	0.44 ± 0.32	3.96 ± 0.70	-1.56 ± 0.12	0.48 ± 0.19	0.63 ± 0.04	4.89 ± 0.32	3.14 ± 0.90	1.50 ± 0.32	14.21 ± 1.88
			final	0.59 ± 0.32	7.57 ± 0.70	-1.19 ± 0.12	0.22 ± 0.19	1.31 ± 0.04	3.31 ± 0.32	4.64 ± 0.90	1.40 ± 0.32	15.06 ± 1.88
			eval	0.73 ± 0.62	8.96 ± 0.63	-1.59 ± 0.71	0.59 ± 0.71	-0.08 ± 0.07	5.67 ± 1.34	3.12 ± 3.68	1.49 ± 1.21	14.50 ± 2.27
		20	trian	-0.00 ± 0.15	8.46 ± 0.92	-1.68 ± 0.37	0.58 ± 0.34	0.50 ± 0.11	6.90 ± 3.90	0.99 ± 1.91	1.36 ± 1.14	15.09 ± 1.10
			final	0.21 ± 0.15	8.46 ± 0.92	-0.82 ± 0.37	-0.19 ± 0.34	-0.54 ± 0.11	5.77 ± 3.92	2.89 ± 1.91	1.49 ± 1.14	16.68 ± 1.10
			eval	0.36 ± 0.44	7.36 ± 2.68	-0.80 ± 0.85	0.75 ± 0.07	0.88 ± 0.23	7.85 ± 3.98	1.41 ± 1.31	0.83 ± 1.62	16.17 ± 1.05
		30	trian	0.71 ± 0.32	6.10 ± 0.62	-1.96 ± 0.20	0.28 ± 0.15	0.43 ± 0.37	-8.66 ± 0.68	0.53 ± 0.90	-0.42 ± 0.18	16.79 ± 0.60
			final	1.29 ± 0.32	8.07 ± 0.62	-1.24 ± 0.20	0.46 ± 0.15	0.47 ± 0.37	-8.55 ± 0.68	4.51 ± 0.90	0.73 ± 0.18	18.10 ± 0.63
			eval	0.20 ± 0.50	8.03 ± 2.33	-1.92 ± 0.41	0.60 ± 0.86	0.67 ± 0.36	-6.82 ± 5.34	4.28 ± 3.57	0.22 ± 0.82	16.40 ± 1.63
	Assist09	10	trian	-3.23 ± 1.08	-2.68 ± 0.68	-3.10 ± 0.45	-5.12 ± 0.17	-4.12 ± 0.29	7.37 ± 1.17	-7.80 ± 9.47	-1.90 ± 1.71	8.92 ± 0.02
			final	-4.42 ± 1.08	-2.82 ± 0.68	-3.82 ± 0.45	-6.48 ± 0.17	-5.28 ± 0.29	7.62 ± 1.17	-5.20 ± 9.47	-0.65 ± 1.71	9.49 ± 0.02
			eval	-2.10 ± 0.70	-2.65 ± 1.54	-3.69 ± 0.63	-4.91 ± 1.75	-4.28 ± 1.09	7.31 ± 1.08	-5.31 ± 13.37	-2.88 ± 1.69	7.95 ± 0.38
		20	trian	-2.72 ± 0.42	-2.10 ± 0.35	-3.54 ± 1.02	-4.97 ± 0.13	-3.26 ± 0.17	7.37 ± 2.24	-1.68 ± 4.78	-1.81 ± 2.34	8.95 ± 0.11
			final	-2.30 ± 0.42	-3.04 ± 0.35	-4.11 ± 1.02	-5.40 ± 0.13	-4.85 ± 0.17	6.97 ± 2.24	-1.43 ± 4.78	-2.81 ± 2.34	9.47 ± 0.11
			eval	-1.27 ± 1.19	-1.89 ± 1.17	-3.31 ± 0.86	-4.25 ± 0.49	-2.66 ± 1.65	7.78 ± 0.62	-0.73 ± 4.17	-4.58 ± 5.92	8.66 ± 0.00
		30	trian	-3.75 ± 0.70	-2.07 ± 0.59	3.05 ± 3.15	-5.14 ± 0.27	-3.74 ± 0.36	8.40 ± 0.63	-4.29 ± 2.82	-2.63 ± 1.07	8.55 ± 0.06
			final	-3.48 ± 0.70	0.90 ± 0.59	-7.80 ± 3.15	-4.86 ± 0.27	-2.32 ± 0.36	7.49 ± 0.63	-0.75 ± 2.82	-2.29 ± 1.07	9.47 ± 0.06
			eval	-1.98 ± 0.92	-1.46 ± 1.01	-1.48 ± 4.84	-4.18 ± 0.81	-2.82 ± 1.08	8.22 ± 0.00	1.90 ± 5.33	-2.68 ± 0.36	8.64 ± 0.02

Table 4: Performance comparison across different learning patterns, datasets, steps, and stages (values are mean±std %).